General Heisenberg model

$$H = \sum_{\langle n,m \rangle} J(\vec{R}_n - \vec{R}_m)\, \vec{S}_n \cdot \vec{S}_m$$

Classical spin configuration: $\vec{S}_R$ = classical vector; $|\vec{S}_R| = S$.

let lattice be a Bravais lattice with one spin per unit cell.

$$J(\vec{R}_n - \vec{R}_m) = J(\vec{R}_m - \vec{R}_n)$$

Fourier: $\vec{S}_q = \frac{1}{N^{1/2}} \sum_n S_n e^{-i\vec{q}\vec{R}_n}$

Lattice vectors $\vec{R}_n = n_1\vec{a}_1 + n_2\vec{a}_2 + n_3\vec{a}_3$ where n_i = integer

Reciprocal lattice vectors $\vec{K}_n = m_1\vec{b}_1 + m_2\vec{b}_2 + m_3\vec{b}_3$

$\Big\}\ e^{i\vec{K}\vec{R}} = 1$

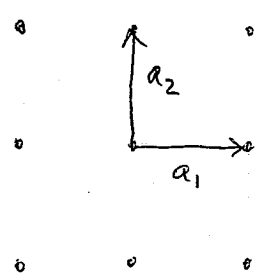
where $\vec{b}_1 = 2\pi \dfrac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3}$ $\quad \vec{b}_2 = \dfrac{2\pi\, \vec{a}_3 \times \vec{a}_1}{\vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3}$ $\quad \vec{b}_3 = \dfrac{2\pi\, \vec{a}_1 \times \vec{a}_2}{\vec{a}_1 \cdot \vec{a}_2 \times \vec{a}_3}$

Write momentum $\vec{q} = q_1\vec{b}_1 + q_2\vec{b}_2 + q_3\vec{b}_3$

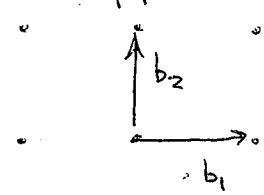
then $q_i = \dfrac{m_i}{N_i}$, m_i = integer, N_i = number of unit cells in $\vec{a}_i$ direction

$\Rightarrow$ the number of independent $\vec{q}$ vectors is $N_1 \cdot N_2 \cdot N_3$ = number of spins in the lattice.

Let square lattice

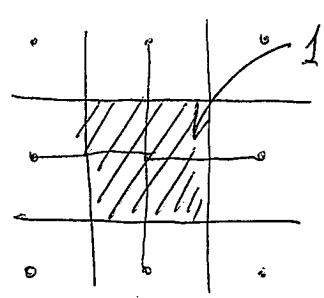


Reciprocal lattice



$b_1 = \dfrac{2\pi}{a_1}$

$b_2 = \dfrac{2\pi}{a_2}$



1st Brillouin zone (BZ)

$$\vec{S}_q = N^{-1/2} \sum_n \vec{S}_n e^{-i\vec{q}\vec{R}_n}$$

$$\vec{S}_{-q} = N^{-1/2} \sum_n \vec{S}_n e^{i\vec{q}\vec{R}_n} = \left(\vec{S}_q\right)^*$$

$$\Rightarrow \vec{S}_n = N^{-1/2} \sum_q \vec{S}_q e^{i\vec{q}\vec{R}_n}$$

Check: $\vec{S}_n = N^{-1/2} \sum_q \left(N^{-1/2} \sum_m S_m e^{-i\vec{q}\vec{R}_m} \right) e^{i\vec{q}\vec{R}_n} =$

$$= \sum_m \underbrace{N^{-1} \sum_q e^{i\vec{q}(\vec{R}_n - \vec{R}_m)}}_{\delta_{n,m}} \vec{S}_m = \vec{S}_n \quad \underline{\text{ok}}$$

Then $H = \frac{1}{2} \sum_{n,m} J(R_n - R_m)\, N^{-1/2} \sum_{q_1} \vec{S}_{q_1} e^{i\vec{q}_1\vec{R}_n}\, N^{-1/2} \sum_{q_2} \vec{S}_{q_2} e^{i\vec{q}_2\vec{R}_m}$

let $\vec{R}_n - \vec{R}_m = \vec{r}$

$$H = \frac{1}{2} \sum_{n,r} J(r)\, N^{-1} \sum_{q_1 q_2} \vec{S}_{q_1} \vec{S}_{q_2}\, e^{i\vec{R}_n(\vec{q}_1 + \vec{q}_2)}\, e^{-i\vec{q}_2\vec{r}}$$

$$\frac{1}{N} \sum_n e^{i\vec{R}_n(\vec{q}_1 + \vec{q}_2)} = \delta_{q_1, -q_2}$$

$$H = \frac{1}{2} \sum_{q_1} \sum_r J(r)\, e^{i\vec{q}_1\vec{r}}\, \vec{S}_{q_1} \cdot \vec{S}_{-q_1} = \sum_q \underbrace{\left(\frac{1}{2} \sum_r J(r) e^{i\vec{q}\vec{r}} \right)}_{J(q)} \vec{S}_q \cdot \vec{S}_{-q}$$

even function

$$J(q) = \frac{1}{2} \sum_r J(r) e^{i\vec{q}\cdot\vec{r}} \underset{\vec{r} = -\vec{t}}{=} \frac{1}{2} \sum_t J(-t) e^{-i\vec{q}\vec{t}} =$$

$$= \frac{1}{2} \sum_t J(t)\, e^{-i\vec{q}\cdot\vec{t}} = J(-q)$$

Square lattice

$$J(q) = \frac{1}{2} J\left(e^{i\vec{q}\cdot\vec{a}_1} + e^{+i\vec{q}\cdot(-\vec{a}_1)} + e^{i\vec{q}\cdot\vec{a}_2} + e^{-i\vec{q}(-\vec{a}_2)} \right)$$

$$= J\left(\cos q_x a + \cos q_y a \right)$$

(1) $$H = \sum_q J(q)\, \vec{S}_q \cdot \vec{S}_{-q} = \sum_q J(q)\, \underbrace{\vec{S}_q \cdot (\vec{S}_q)^*}_{|S_q^x|^2 + |S_q^y|^2 + |S_q^z|^2}$$

positive!

Clearly minimum of energy will be obtained when we choose $\vec{Q}$ such that $J(Q) =$ minimal possible value.

But there is <u>constraint</u>: on every site the magnitude of spin is fixed!

(C1) $$\vec{S}_n \cdot \vec{S}_n = S^2 = N^{-1} \sum_{qq'} \vec{S}_q \cdot \vec{S}_{q'}\, e^{i\vec{q}\vec{R}_n + i\vec{q}'\vec{R}_n}$$

In general, constraint (C1) is pretty difficult to handle — have to minimize (1) subject to (C1).

Instead, consider <u>relaxed constraint</u> (where we average over all sites)

(C2) $$\sum_n \vec{S}_n \cdot \vec{S}_n = \sum_n S^2 = N S^2$$

$$\Rightarrow \quad N S^2 = \sum_n \frac{1}{N} \sum_{qq'} \vec{S}_q \cdot \vec{S}_{q'}\, e^{i\vec{R}_n(\vec{q}+\vec{q}')} = \sum_q \vec{S}_q \cdot \vec{S}_{-q}$$

$\ast$ The strategy is then to minimize (1) subject to (C2), and then check if so found $\vec{Q}$ satisfies (C1). If it does, than the full problem is solved.

Solution: $\vec{S}_q = 0$ for all $\vec{q}$ except $\vec{q} = \vec{Q}$

$$\vec{S}_Q \neq 0$$

where $\vec{Q}$ minimizes $J(Q)$. In fact, since $J(q) = J(-q)$, both $\pm\vec{Q}$ should be taken

Then (C2) $$NS^2 = \sum_{\substack{q=Q \\ q=-Q}} \vec{S}_q \cdot \vec{S}_{-q} = 2\,\vec{S}_Q \cdot \vec{S}_{-Q}$$

while (C1) $$S^2 = \frac{1}{N} \sum_{\substack{q=\pm Q \\ q'=\pm Q}} \vec{S}_q \cdot \vec{S}_{q'}\, e^{i\vec{R}_n(\vec{q}+\vec{q}')} =$$

$$= \frac{1}{N}\Big\{ \vec{S}_Q \cdot \vec{S}_Q\, e^{i2\vec{Q}\cdot\vec{R}_n} + \vec{S}_{-Q}\cdot\vec{S}_{-Q}\, e^{-i2\vec{Q}\vec{R}_n}$$

$$+ \vec{S}_Q \cdot \vec{S}_{-Q}\, \underbrace{e^{i\vec{R}_n(\vec{Q}-\vec{Q})}}_{1} + \vec{S}_{-Q}\cdot\vec{S}_Q \Big\}$$

But the left-hand side is independent of $\vec{R}_n \Rightarrow$

$$\vec{S}_Q \cdot \vec{S}_Q = 0, \quad \vec{S}_{-Q}\cdot\vec{S}_{-Q} = 0$$

Introduce <u>real vectors</u> $\vec{N}_Q, \vec{M}_Q$:

$$\vec{S}_Q = \vec{N}_Q + i\vec{M}_Q, \quad \vec{S}_{-Q} = \vec{N}_Q - i\vec{M}_Q$$

$$\Rightarrow \vec{S}_Q \cdot \vec{S}_Q = \vec{N}_Q\cdot\vec{N}_Q - \vec{M}_Q\cdot\vec{M}_Q + 2i\,\vec{N}_Q\cdot\vec{M}_Q = 0$$

$$\Rightarrow \underbrace{\vec{N}_Q\cdot\vec{N}_Q = \vec{M}_Q\cdot\vec{M}_Q}_{\text{equal magnitude}} \quad \text{and} \quad \underbrace{\vec{N}_Q\cdot\vec{M}_Q = 0}_{\text{orthogonal vectors}}$$

Thus $$NS^2 = 2\,\vec{S}_Q\cdot\vec{S}_{-Q} = 2\left(\vec{N}_Q + i\vec{M}_Q\right)\left(\vec{N}_Q - i\vec{M}_Q\right)$$

$$= 2\left(\vec{N}_Q^2 + \vec{M}_Q^2\right) = 4\,\vec{N}_Q^2 = 4\,\vec{M}_Q^2$$

$$\Rightarrow \underline{\vec{N}_Q^2 = \vec{M}_Q^2 = \frac{1}{4}NS^2} \Rightarrow \text{choose } \vec{N}_Q = \vec{n}_1\sqrt{\frac{NS^2}{4}}$$

$$\vec{M}_Q = \vec{n}_2\sqrt{\frac{NS^2}{4}}$$

$$\vec{n}_1\cdot\vec{n}_2 = 0$$

The minimal energy $H_{min} = \sum_{q=\pm Q} J(q)\, \vec{S}_q \cdot \vec{S}_{-q} = 2J(Q)\, \vec{S}_Q \cdot \vec{S}_{-Q}$

$$= J(Q)\, N S^2 .$$

$\Rightarrow$ energy per site $= \underline{J(Q) S^2}$.

$\Rightarrow$ Real space structure: $\vec{S}_n = N^{-1/2} \sum_{q=\pm Q} \vec{S}_q e^{i\vec{q}\vec{R}_n} =$

$$= N^{-1/2}\left(\vec{S}_Q e^{i\vec{Q}\vec{R}_n} + \vec{S}_{-Q} e^{-i\vec{Q}\vec{R}_n}\right) =$$

$$= N^{-1/2}\left(\vec{N}_Q \underbrace{\left(e^{i\vec{Q}\vec{R}_n} + e^{-i\vec{Q}\vec{R}_n}\right)}_{2\cos(\vec{Q}\cdot\vec{R}_n)} + i\vec{M}_Q \underbrace{\left(e^{i\vec{Q}\vec{R}_n} - e^{-i\vec{Q}\vec{R}_n}\right)}_{2i\sin(\vec{Q}\cdot\vec{R}_n)}\right) =$$

$$= \frac{2}{\sqrt{N}}\left\{\vec{N}_Q \cos(\vec{Q}\cdot\vec{R}_n) - \vec{M}_Q \sin(\vec{Q}\cdot\vec{R}_n)\right\} =$$

$$= S\left\{\vec{n}_1 \cos(\vec{Q}\cdot\vec{R}_n) - \vec{n}_2 \sin(\vec{Q}\cdot\vec{R}_n)\right\}$$

where $\vec{n}_1^{\,2} = \vec{n}_2^{\,2} = 1$ and $\vec{n}_1 \cdot \vec{n}_2 = 0$ } helical (spiral) spin structure

For a given $\vec{Q}$, which minimizes $J(Q)$, choose a plane perpendicular to $\vec{Q}$ — on that plane $\vec{Q}\cdot\vec{R}_n = \text{const}$. Within such a plane all spins are parallel to each other. Choosing another $\vec{R}_n'$: $\vec{Q}\cdot\vec{R}_n' =$ different constant. Spins in this plane are also parallel amongst themselves and make angle $\vec{Q}\cdot(\vec{R}_n' - \vec{R}_n)$ with spins from $\vec{R}_n$ plane.

For example, $\vec{n}_1 = \hat{x}$, $\vec{n}_2 = \hat{y}$ $\Rightarrow$

$$\vec{S}_n = S\left(\cos(\vec{Q}\cdot\vec{R}_n + \theta),\ S\sin(\vec{Q}\cdot\vec{R}_n + \theta),\ 0\right)$$

Energy does not depend on θ.

<u>Examples</u> ① square lattice ferromagnet

$$J(r) = \begin{cases} -J, & \text{when } \vec{r} = \text{nearest neighbors} \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow J(q) = -J\left(\cos q_x a + \cos q_y a\right)$$

Minimized by $\vec{Q} = (0,0)$

$$J(\vec{Q}) = -2J$$

$$\begin{array}{|ccc|} \uparrow & \uparrow & \uparrow \\ \uparrow & \uparrow & \uparrow \end{array}$$

<u>FM</u> $\vec{S}_n = S\left(\cos\theta, \sin\theta, 0\right)$ = constant vector on every site

② square lattice antiferromagnet

$$J(r) = \begin{cases} J, & \vec{r} = \text{nearest neighbors} \\ 0, & \text{otherwise} \end{cases}$$

$$J(q) = J\left(\cos q_x a + \cos q_y a\right)$$

Minimized by $\vec{Q} = \left(\frac{\pi}{a}, \frac{\pi}{a}\right)$: $J(Q) = -2J$

belongs to BZ boundary!

$$\vec{Q}\cdot\vec{R}_n = \pi n_1 + \pi n_2 = \pi(n_1+n_2)$$

$\left(\frac{\pi}{a}\hat{x} + \frac{\pi}{a}\hat{y}\right)$ $\quad \hookrightarrow (n_1 a\hat{x} + n_2 a\hat{y})$

$$\begin{array}{|ccc|} \uparrow & \downarrow & \uparrow \\ \downarrow & \uparrow & \downarrow \end{array}$$

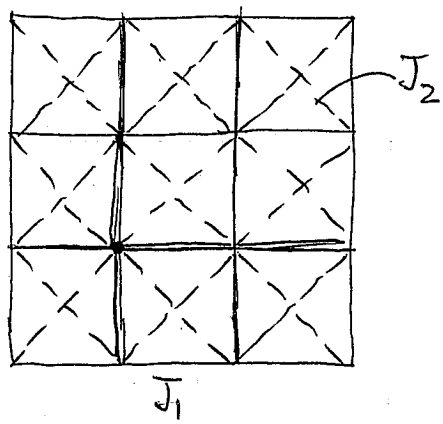
$$\vec{S}_n = S\left(\cos\left(\pi(n_1+n_2)+\theta\right), \sin\left(\pi(n_1+n_2)+\theta\right), 0\right)$$

For simplicity, choose direction $(\cos\theta, \sin\theta, 0)$ as x-axis $\Rightarrow$ this means "set $\theta = 0$ on the expression above".

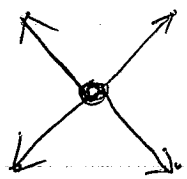
<u>AFM</u> $\vec{S}_n = S\left(\cos\pi(n_1+n_2), \underbrace{\sin\pi(n_1+n_2)}_{\emptyset}, 0\right) = S\left((-1)^{n_1+n_2}, 0, 0\right)$

(3) $J_1 - J_2$ model (solid lines J_1, dashed lines J_2)

$J_1 > 0$, $J_2 > 0$



$$J(q) = J_1(\cos q_x a + \cos q_y a) + \tfrac{1}{2} J_2 \left(e^{i(q_x+q_y)a} + e^{-i(q_x+q_y)a} + e^{i(q_x-q_y)a} + e^{-i(q_x-q_y)a} \right)$$



$$\cos(q_x+q_y)a + \cos(q_x-q_y)a = 2\cos(q_x a)\cos(q_y a)$$

$$J(q) = J_1\left(\cos(q_x a) + \cos(q_y a)\right) + 2J_2 \cos(q_x a)\cos(q_y a) =$$

$$= J_1 \left\{ \cos\alpha + \cos\beta + \mu \cos\alpha \cos\beta \right\}$$

Notations: $\alpha = q_x a$, $\beta = q_y a$, $\mu = \dfrac{2J_2}{J_1}$

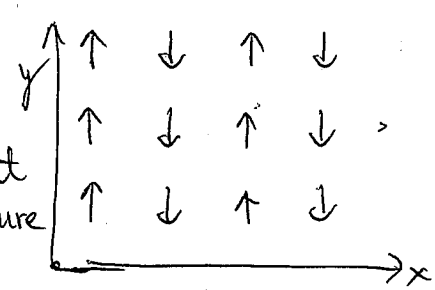
Simple limits: (A) $\mu \to 0 \Rightarrow$ then $\alpha = \beta = \pi$ $\quad J(Q) = -2J_1$ AFM state ($J_2 \ll J_1$)

(B) $\mu \to \infty \Rightarrow$ then $\begin{cases} \alpha = \pi, \beta = 0 \\ \alpha = 0, \beta = \pi \end{cases}$ "4-sublattice" state $\quad J(Q) = -2J_2$

Suppose $q_x = \frac{\pi}{a}$, $q_y = 0$: $\vec{S}_n = S(\cos \pi n_1, \sin \pi n_1, 0) = S((-1)^{n_1}, 0, 0)$

$\vec{Q} \cdot \vec{R}_n = q_x n_1 a = \pi n_1$

y-independent spin structure



lines of up down spins at even, odd x

(c) Special case $\mu = 1$: $\underline{J_2 = \frac{1}{2} J_1}$

$$\frac{J(q)}{J_1} = \cos\alpha + \cos\beta + \cos\alpha\cos\beta = \cos\alpha \underbrace{(1+\cos\beta)}_{2\cos^2\frac{\beta}{2}} + \underbrace{\cos\beta}_{2\cos^2\frac{\beta}{2}-1} =$$

$$= -1 + 2\cos^2\frac{\beta}{2}(1+\cos\alpha)$$

$$J(q)/J_1 = -1 + 4\cos^2\left(\frac{\alpha}{2}\right)\cos^2\left(\frac{\beta}{2}\right)$$

$$J(q) = J_1\left\{-1 + 4\cos^2\left(\frac{q_x a}{2}\right)\cos^2\left(\frac{q_y a}{2}\right)\right\}$$

Minimum of energy when the last term $=0$
(for otherwise it is always positive!)

$\Rightarrow$ <u>lines of zeroes</u> : $\cos^2\left(\frac{q_x a}{2}\right)\cos^2\left(\frac{q_y a}{2}\right) = 0$

$q_x = \frac{\pi}{a}$, arbitrary q_y !

OR q_x = arbitrary, $q_y = \frac{\pi}{a}$.

let $\vec{Q} = \left(\frac{\pi}{a}, q_y\right)$, $J(Q) = -J_1$

$$\vec{Q}\cdot\vec{R}_n = \pi n_1 + \underbrace{q_y a}_{\beta = \text{arbitrary}} n_2 = \pi n_1 + \beta n_2$$

$$\vec{S}_n = S\left(\cos(\pi n_1 + \beta n_2),\ \sin(\pi n_1 + \beta n_2),\ 0\right) =$$

$$= S\left((-1)^{n_1}\cos(\beta n_2),\ (-1)^{n_1}\sin(\beta n_2),\ 0\right)$$

<u>Classical degeneracy</u>: states with different β have the same energy!

$\boxed{?}$

$$H = \frac{1}{2} \sum_{n,m} J(R_n - R_m)\, \vec{S}_n \cdot \vec{S}_m$$

$$\vec{S}_n = \underbrace{\langle \vec{S}_n \rangle}_{\text{average}} + \underbrace{\vec{S}_n - \langle \vec{S}_n \rangle}_{\text{fluctuations} = \delta \vec{S}_n}$$

$$H \approx \frac{1}{2} \sum_{n,m} J(R_n - R_m) \left(\langle \vec{S}_n \rangle + \delta \vec{S}_n \right) \left(\langle S_m \rangle + \delta \vec{S}_m \right)$$

$$\approx \frac{1}{2} \sum_{n,m} J(R_n - R_m) \langle \vec{S}_n \rangle \langle \vec{S}_m \rangle + \frac{1}{2} \sum_{nm} J(R_n - R_m) \langle \vec{S}_n \rangle \cdot \left(\vec{S}_m - \langle \vec{S}_m \rangle \right)$$

$$+ \frac{1}{2} \sum J(R_n - R_m) \langle \vec{S}_m \rangle \cdot \left(\vec{S}_n - \langle \vec{S}_n \rangle \right) +$$

$$+ \frac{1}{2} \sum J(R_n - R_m) \underbrace{\delta \vec{S}_n \cdot \delta \vec{S}_m}_{\text{neglect as small}} =$$

$$= -\frac{1}{2} \sum_{nm} J(R_n - R_m) \langle \vec{S}_n \rangle \langle \vec{S}_m \rangle + \sum_{nm} J(R_n - R_m) \langle \vec{S}_m \rangle \vec{S}_n$$

$$= \text{const} + \sum_n \vec{S}_n \cdot \underbrace{\left(\sum_m J(R_n - R_m) \langle \vec{S}_m \rangle \right)}_{-g\mu_B \vec{H}_{eff}}$$

$$\langle \vec{S}_m \rangle = \sigma \left(\cos(\vec{Q} \cdot \vec{R}_m),\ \sin \vec{Q} \cdot \vec{R}_m,\ 0 \right)$$

↳ magnitude of the average of the spin

$$\sum_m J(R_n - R_m) \langle \vec{S}_m \rangle = \sigma \sum_m J(R_n - R_m) \left[\hat{x} \cos(\vec{Q} \vec{R}_m) + \hat{y} \sin \vec{Q} \vec{R}_m \right]$$

But $\sum_m J(R_m - R_n)\, e^{i\vec{Q}\vec{R}_m} = \sum_m J(R_m - R_n)\, e^{iQ(\vec{R}_m - \vec{R}_n)}\, e^{i\vec{Q}\vec{R}_n}$

$$= \underbrace{\sum_r J(r)\, e^{iQr}}_{2J(Q)}\, e^{i\vec{Q}\vec{R}_n} = 2J(Q)\, e^{i\vec{Q}\vec{R}_n}$$

$$\Rightarrow \sum_m J(R_m - R_n) \langle \vec{S}_m \rangle = \sigma J(Q) \left(\cos(\vec{Q}\vec{R}_n)\, \hat{x} + \sin(\vec{Q}\vec{R}_n)\, \hat{y} \right)$$

$$= -g\mu_B \vec{H}_{eff}$$

$\Rightarrow$ The field experienced by spin $\vec{S}_n$ is self-consistent in that it coincides with the direction of $\langle \vec{S}_n \rangle = \sigma \left(\cos \vec{Q}\vec{R}_n,\ \sin \vec{Q}\vec{R}_n,\ 0 \right)$!

The magnitude of that field is $|\vec{H}_{eff}| = \dfrac{\sigma J(Q)}{g\mu_B}$.

As before, we determine it self-consistently:

$$\frac{|\langle S_n \rangle|}{S} = \frac{\sigma}{S} = \frac{\sum_{m_s=-S}^{S} \frac{m_s}{S} \exp\left(\frac{\sigma J(Q) S}{k_B T} \frac{m_s}{S} \right)}{\sum_{m_s} \exp\left(\frac{\sigma J(Q) S}{k_B T} \frac{m_s}{S} \right)} \Rightarrow$$

$$\Rightarrow \frac{\sigma}{S} = B_S\left(\frac{\sigma J(Q) S}{k_B T} \right).$$

Brillouin function again.

$$B_S(x) = \frac{2S+1}{2S}\coth\frac{2S+1}{2S}x - \frac{1}{2S}\coth\frac{1}{2S}x$$

$$(x\to 0) \approx \frac{S+1}{3S}x - \underbrace{\frac{[(S+1)^2+S^2](S+1)}{90S^3}}_{=\gamma}x^3 + \dots$$

Thus

$$\frac{\sigma}{S} = \frac{S+1}{3S}\frac{J(Q)S\sigma}{k_BT} - \gamma\left(\frac{J(Q)S\sigma}{k_BT}\right)^3$$

(1) $\sigma \neq 0$ solution appears at

$$T_Q = \frac{S(S+1)J(Q)}{3k_B}$$

(2) near that T

$$1 = \frac{T_Q}{T} - \gamma S\left(\frac{J(Q)S}{k_BT}\right)^3\sigma^2$$

$$\gamma S\left(\frac{J(Q)S}{k_BT}\right)^3\sigma^2 = \frac{T_Q}{T} - 1;$$

For $T \approx T_c$ we can approximate $\frac{J(Q)S}{k_BT} \approx \frac{J(Q)S}{k_BT_Q} =$

$$= \frac{J(Q)S}{k_B\frac{S(S+1)J(Q)}{3k_B}} = \frac{3S}{S(S+1)}$$

$$\Rightarrow \sigma^2 = \left(\frac{S(S+1)}{3S}\right)^3\frac{1}{\gamma S}\frac{T_Q - T}{T_Q}$$

But $\gamma = \dfrac{\left[(S+1)^2 + S^2\right](S+1)}{90S^3}$

$$\Rightarrow \quad \sigma^2 = \left(\frac{\cancel{S}(S+1)}{3\cancel{S}}\right)^3 \frac{90S^2}{\left[(S+1)^2+S^2\right](S+1)} \quad \frac{T_Q - T}{T}$$

$$= \frac{10\,S^2(S+1)^2}{(S+1)^2+S^2} \quad \frac{T_Q - T}{T}$$

Importantly, $\sigma \sim \sqrt{\dfrac{T_Q}{T} - 1}$ for $T \leq T_Q$

<u>Finite-q susceptibility</u>

Let us apply modulated magnetic field

$$\vec{B}(r) = \vec{B}\cos(\vec{q}\cdot\vec{r})$$

$$H = \sum_q J(q)\,\vec{S}_q\cdot\vec{S}_{-q} - g\mu_B \underbrace{\sum_R \vec{B}(R)\cdot\vec{S}_R}$$

$$\frac{1}{2}\sum_R \vec{B}\cdot\vec{S}_R\left(e^{i\vec{q}\vec{r}} + e^{-i\vec{q}\vec{r}}\right) =$$

$$= \frac{1}{2}\vec{B}\cdot N^{1/2}\left(\vec{S}_q + \vec{S}_{-q}\right)$$

Above critical temperature the only component that will be induced by the field will be one with the same momenta, $\vec{q}$ and $-\vec{q}$:

$$H \approx J(q)\langle\vec{S}_q\rangle\vec{S}_{-q} + J(q)\langle\vec{S}_{-q}\rangle\vec{S}_q - g\frac{\mu_B\sqrt{N}}{2}B\cdot(\vec{S}_q+\vec{S}_{-q})$$

magnetization at $\vec{q}$

$$\vec{M}(q) = g\mu_B\langle\vec{S}_q\rangle = \chi_0\vec{H}_{eff}(-q)$$

↳ single spin susceptibility from Curie law: $\chi_0 = \frac{C}{T}$

$$C = g^2\mu_B^2\frac{S(S+1)}{3k_B}$$

$$\vec{H}_{eff}(-q) = \left(\text{field acting on } \vec{S}_q\right) = \frac{\sqrt{N}}{2}\vec{B} - \frac{J(q)\langle\vec{S}_{-q}\rangle}{g\mu_B}$$

Thus $$g\mu_B\langle\vec{S}_q\rangle = \chi_0\left(\frac{\sqrt{N}}{2}\vec{B} - \frac{J(q)\langle\vec{S}_q\rangle}{g\mu_B}\right)$$

$$\Rightarrow \left(g\mu_B + \frac{\chi_0 J(q)}{g\mu_B}\right)\langle\vec{S}_q\rangle = \chi_0\frac{\sqrt{N}}{2}\vec{B}$$

On the other hand, $\vec{M}(q) = \chi(q)\frac{\sqrt{N}}{2}\vec{B}$

↳ susceptibility in responce to applied field $\vec{B}$.

$$\chi(q)\frac{\sqrt{N}}{2}\vec{B} = g\mu_B\frac{\chi_0\frac{\sqrt{N}}{2}\vec{B}}{g\mu_B + \frac{\chi_0 J(q)}{g\mu_B}}$$

$$\chi(q) = \frac{\chi_0}{1 + \frac{\chi_0 J(q)}{g^2\mu_B^2}} = \frac{C/T}{1 + \frac{C J(q)}{g^2\mu_B^2 T}} =$$

$$= \frac{C}{T + \frac{C J(q)}{g^2\mu_B^2}} = \frac{C}{T + T_Q \frac{J(q)}{J(Q)}}$$

But $T_Q = \frac{J(Q)\, S(S+1)}{3 k_B}$ and $\frac{C}{g^2\mu_B^2} = \frac{S(S+1)}{3k_B}$

<u>FM</u> $T_0 = \frac{Z J S(S+1)}{2 \cdot 3 k_B}$, $J(q=0) = -\frac{Z}{2} J$

$$\chi(q=0) = \frac{C}{T - \frac{ZJ}{2} \frac{S(S+1)}{3k_B}} = \frac{C}{T - T_0}$$

<u>AFM</u> $J(0) = \frac{Z J}{2}$

$$\chi(q=0) = \frac{C}{T + \frac{ZJ}{2} \frac{S(S+1)}{3k_B}} = \frac{C}{T + T_Q}$$

$$T_Q = \frac{S(S+1)}{3k_B} |J(Q)| = \frac{ZJ}{2} \frac{S(S+1)}{3k_B}$$

But if we probe at $\vec{q} = \vec{Q}$, then $J(Q) = J(\cos Q_x a + \cos Q_y a)$
$= -\frac{ZJ}{2}$

$$\Rightarrow \chi(Q) = \frac{C}{T - \frac{S(S+1)}{3k_B} \cdot \frac{ZJ}{2}}$$

Susceptibility diverges at the ordering momentum $\vec{Q}$ as T is lowered!